

Assignment 8: Build and Evaluate Time Series Models

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DDS-8555 Predictive Analysis

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```
In [1]: from pathlib import Path
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

ROOT = Path.cwd()
for parent in [ROOT, *ROOT.parents]:
    if (parent / "DDS-8555 - Predictive Analysis").exists():
        COURSE = parent / "DDS-8555 - Predictive Analysis"
        break
else:
    COURSE = ROOT.parents[1]
DATA = COURSE / "data"
KAGGLE = DATA / "kaggle"
SUBMISSIONS = DATA / "submissions"
RANDOM_STATE = 42
pd.set_option("display.max_columns", 80)
```

Signature Assignment Overview

This notebook brings together three competition tasks: Store Sales time series forecasting, House Prices regression, and San Francisco crime classification. The notebook documents the modeling strategy, local diagnostics, generated submission files, and Kaggle evidence. The modeling scope follows the rubric: ETS and ARIMA for Store Sales, two regression models for House Prices, and two tree models plus one SVM for San Francisco Crime.

Store Sales: ETS and ARIMA

The Store Sales forecasts were built at the product-family level and then allocated back to store-family rows using recent sales share. ETS represents level, trend, and seasonality through exponential smoothing, while ARIMA represents autocorrelation and differencing structure (Hyndman & Athanasopoulos, 2021).

```
In [2]: from pathlib import Path
status_path = SUBMISSIONS / "kaggle_submission_status_store-sales-time-series-forec
```

```
text = status_path.read_text(encoding="utf-8", errors="ignore")
print("\n".join([line for line in text.splitlines() if "A8_store_" in line or "file
```

House Prices: Regression with Transformations and Dimension Reduction

The House Prices submissions used regression pipelines that handle dichotomous variables, categorical encodings, polynomial terms, interactions, transformations, and shrinkage. Ridge regression gives a stable linear benchmark, while gradient boosting captures nonlinearities and interactions more directly.

```
In [3]: from pathlib import Path
status_path = SUBMISSIONS / "kaggle_submission_status_house-prices-advanced-regress
text = status_path.read_text(encoding="utf-8", errors="ignore")
print("\n".join([line for line in text.splitlines() if "A8_house_" in line or "file
```

San Francisco Crime: Trees and SVM

The San Francisco crime task is multiclass classification with strong time and location structure. The decision tree is interpretable but weak, the random forest improves stability and class probability estimates, and the linear SVM provides a margin-based comparison.

```
In [4]: from pathlib import Path
status_path = SUBMISSIONS / "kaggle_submission_status_sf-crime.txt"
text = status_path.read_text(encoding="utf-8", errors="ignore")
print("\n".join([line for line in text.splitlines() if "A8_sf_" in line or "fileNam
```

```
In [5]: summary_files = sorted(SUBMISSIONS.glob("A8_*.csv"))
pd.DataFrame({
    "submission_file": [p.name for p in summary_files],
    "size_mb": [round(p.stat().st_size / 1_000_000, 2) for p in summary_files],
})
```

```
Out[5]:
```

	submission_file	size_mb
0	A8_house_gradient_boosting_house_prices.csv	0.04
1	A8_house_ridge_house_prices.csv	0.04
2	A8_sf_decision_tree_sf_crime.csv	335.07
3	A8_sf_linear_svm_sf_crime.csv	717.21
4	A8_sf_random_forest_sf_crime.csv	697.99
5	A8_store_arima_store_sales.csv	0.79
6	A8_store_ets_store_sales.csv	0.77

Interpretation

The Store Sales leaderboard favored the ETS submission over ARIMA, suggesting that smoothed level and seasonality were more effective than the simple ARIMA specification used here. In House Prices, gradient boosting scored better than ridge, which is consistent with a data set that contains nonlinearities, thresholds, and interactions. In SF Crime, the random forest clearly outperformed the single decision tree and linear SVM on log loss, which is expected because random forests can smooth probability estimates across many trees while still using time, district, and coordinate information.

References

Breiman, L. (2001). Random forests. *Machine Learning*, 45, 5-32.

<https://doi.org/10.1023/A:1010933404324>

Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and Practice* (3rd ed.). OTexts. <https://otexts.com/fpp3/>

Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B*, 58(1), 267-288. <https://doi.org/10.1111/j.2517-6161.1996.tb02080.x>